Dual MOT Apparatus for Quantum Degenerate Gases

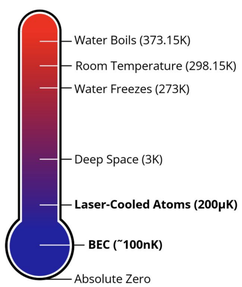
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Abstract

When cooled to ultracold temperatures (T < 1 mK), atomic gases begin to exhibit their wave-like nature due to quantum physics. When bosons are cooled further into the nK-regime, the individual atomic wave functions overlap creating a Bose-Einstein condensate (BEC)- a quantum degenerate gas. Their coherent nature and tunability with external fields make these macroscopic matter waves ideally suited for applications to quantum computation, rotational and gravitational sensors, and precision measurements, to name a few. We discuss our dual magneto-optical trap (MOT) apparatus to achieve BEC. Ultracold samples initially formed in a 2D MOT are transferred from a source chamber to a 3D MOT in the science chamber at ultrahigh vacuum (~ 10^-11 Torr) via a push beam. 3D MOT atoms are detected and characterized through both fluorescence and absorption imaging. We will discuss future work to create a BEC through all-optical trapping and evaporative cooling.

What is Ultracold?



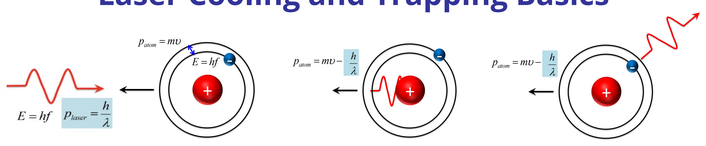
At room temperature, atoms travel at approximately 350 m/s, As the temperature is decreased, the speed of the atoms is reduced.

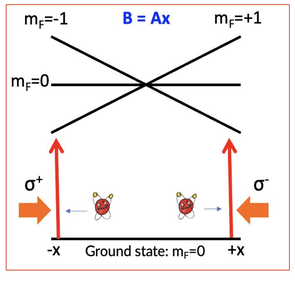
$$\frac{1}{2}mv^2 = \frac{3}{2}k_bT$$ **Slowing = Cooling!**

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2\pi m k_B T}}$$

Lower Temp → Increased Wavelength
Quantum Behavior!

Laser Cooling and Trapping Basics

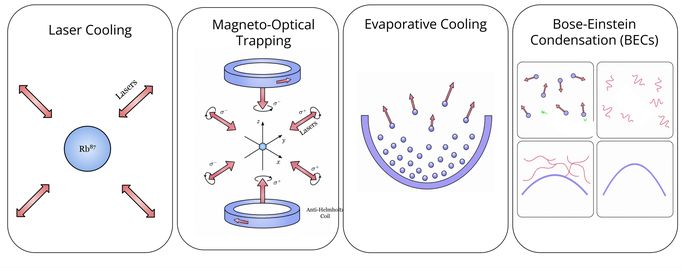




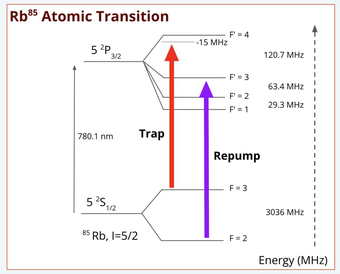
A pair of current-carrying coils creates a magnetic field to shift internal atomic energy levels, allowing an atom to absorb a photon during a collision.

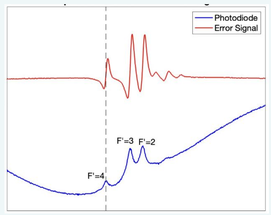
Three pairs of laser beams cool room-temperature atoms to 100 μK (-459 °F!).

Pathway to Bose-Einstein Condensation

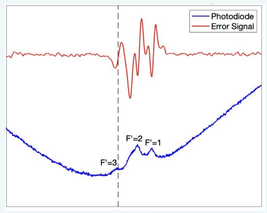


Saturated Absorption Spectroscopy





^{85}Rb Trap

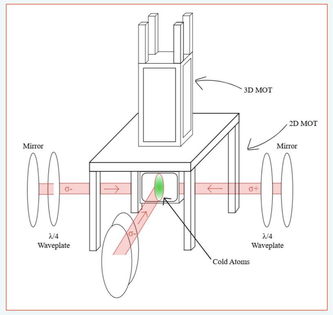


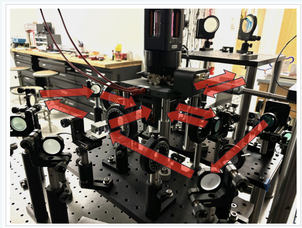
^{85}Rb Repump

The resonance of each transition is represented by a zero crossing in our error signal.

Reducing the scan of the servo controller zooms into the zero crossing of the desired transition and "locks on" to the frequency.

Source Chamber 2D MOT





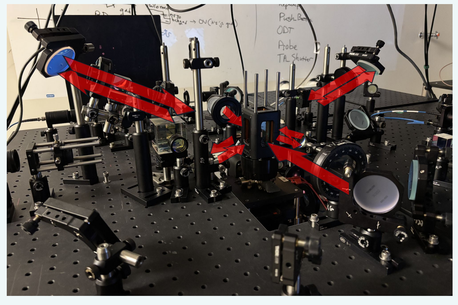
Source Chamber

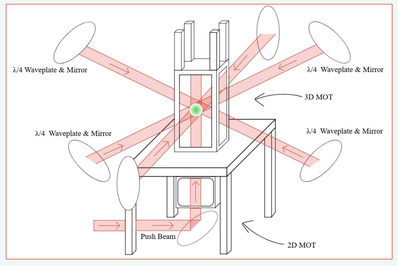
- 85Rb and 87Rb getter source
- Operated at 10^{-9} Torr
- 2D MOT to collect large numbers of ultracold atoms
- Push beam transfers atoms to science chamber (3D MOT)

Science Chamber

- 3D MOT to collect ultracold samples
- Ultrahigh vacuum at 10^{-11} Torr for long lifetimes
- Optical access for dipole trap laser beam and atom detection

Science Chamber 3D MOT

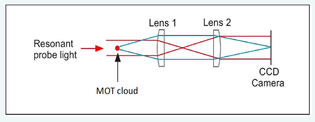




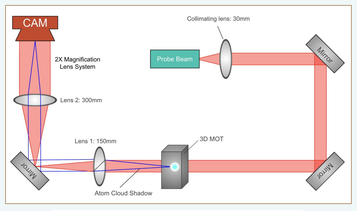


Atom Cloud in 3D MOT Chamber

Absorption Imaging

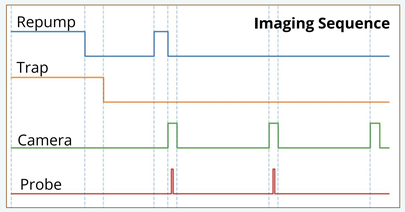


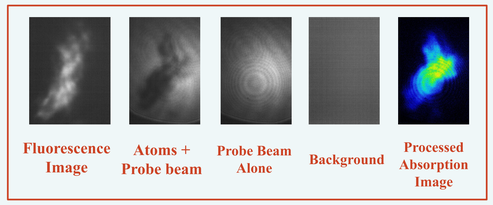
By absorption imaging method, we can also measure the temperature of the atom cloud and the lifetime of the ultracold atoms!



Optical Density:

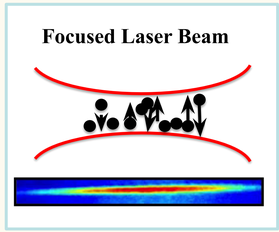
$$N \sim OD = ln\frac{I_{atoms} - I_{bg}}{I_{probe} - I_{bg}}$$

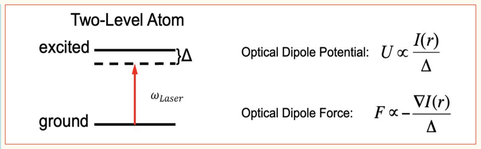


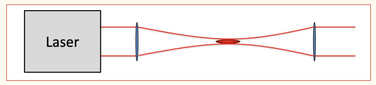


Outlook

Optical Dipole Trap (ODT)

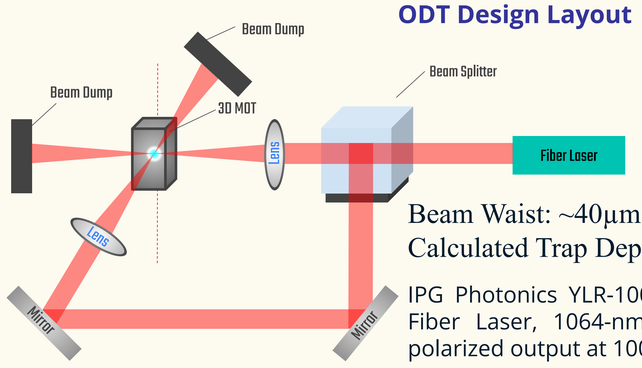






- A focused laser beam traps ultracold atoms through the optical dipole force.
- Lowering the trap light intensity allows hot atoms to evaporate out of the trap leaving the remaining atoms to rethermalize at a lower temperature.
- At 100 nK, the individual atomic wave functions overlap, creating a BEC.

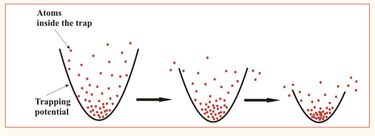
ODT Design Layout



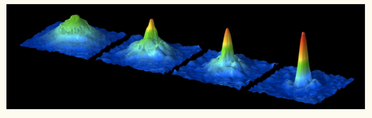
Beam Waist: ~40μm
Calculated Trap Depth: ~2mK

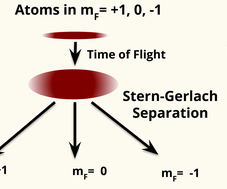
IPG Photonics YLR-100-1064-LP Fiber Laser, 1064-nm, linearly polarized output at 100W

Evaporative Cooling

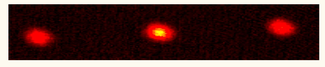


Achieve BEC!





Study of Coherent Spin-Changing Collisions of BEC and Ultracold atoms



Acknowledgements

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